RAK5146 WisLink LPWAN Concentrator Datasheet

Overview

Product Description

The **RAK5146** is an LPWAN Concentrator Module in a mini-PCIe form factor, featuring the Semtech SX1303 and SX126X for the Listen Before Talk (LBT) feature, enabling seamless integration into existing routers or other network equipment with LPWAN Gateway capabilities.

This concentrator can be used with any embedded platform that has an available mini-PCIe slot with SPI/USB connectivity. Additionally, it includes an onboard **ZOE-M8Q GPS** chip.

This module offers an exceptional, comprehensive, and cost-efficient gateway solution. It provides up to 10 programmable parallel demodulation paths, an 8 × 8 channel LoRa packet detector, 8 x SF5-SF12 LoRa demodulators, and 8 x SF5-SF10 LoRa demodulators.

Additionally, RAK5146 can detect an uninterrupted combination of packets at 8 different spreading factors and 10 channels. It can continuously demodulate up to 16 packets, making it suitable for smart metering fixed networks and Internet-of-Things (IoT) applications.

Product Features

- Designed based on **Mini PCI-e form factor**
- **SX1303 baseband processor** emulates 8 × 8 channels LoRa packet detectors, 8x SF5-SF12 LoRa demodulators, 8x SF5-SF10 LoRa demodulators, one 125/250/500 kHz high-speed LoRa demodulator, and one (G)FSK demodulator
- 3.3 V **Mini PCI-e**, compatible with **3G/LTE card** of Mini PCI-e type
- Tx power up to 27 dBm, Rx sensitivity down to -139 dBm @ SF12, BW 125 kHz
- Supports **global license-free frequency band**: EU868, EU433, RU864, CN470, US915, AS923, AU915, KR920, and IN865

- Supports optional **SPI/USB** interfaces.
- Listen Before Talk
- Fine Timestamp
- Built-in **ZOE-M8Q** GPS module

Specifications

Overview

The overview shows the top and back views of the RAK5146 board. It also presents the block diagram that discusses how the board works.

Board Overview

The RAK5146 is a compact LPWAN Gateway Module, making it suitable for integration in systems where mass and size constraints are essential. It has been designed with the PCI Express Mini Card form factor in mind, so it can easily become a part of products that comply with the standard, where they allow cards with a thickness of at least 5.5 mm.

The board has two UFL interfaces for the LoRa and GNSS antennas and a standard 52-pin connector (mPCIe).

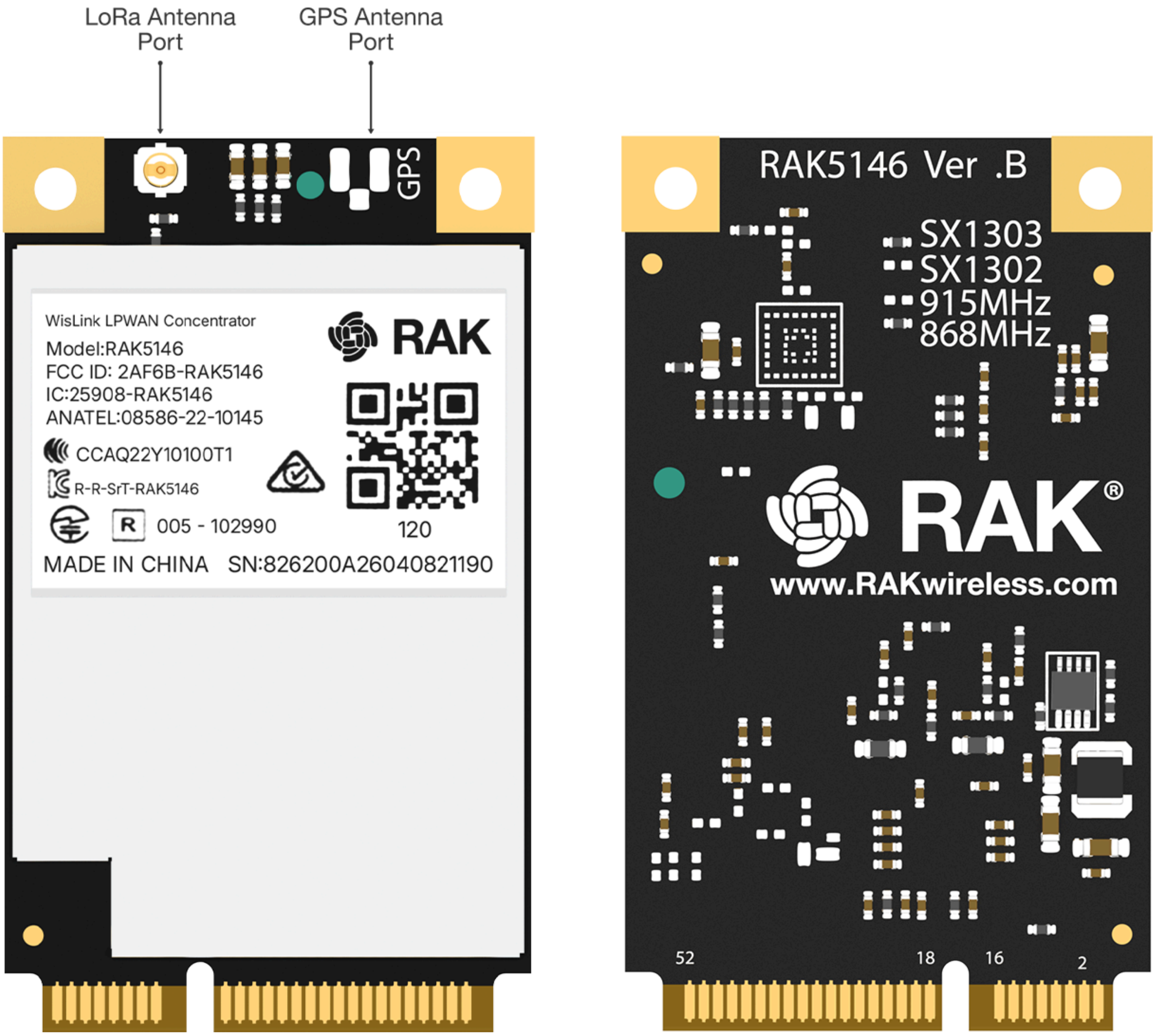


Figure 1: Board Overview

Block Diagram

The RAK5146 concentrator is equipped with one SX1303 chip and two SX1250 chips. One SX1303 chip is used for RF signal processing and serves as the core of the device, while the other SX1250 chip handles the related LoRa modem and processing functionalities.

Additional signal conditioning circuitry is implemented to comply with the PCI Express Mini Card standard, and one UFL connector is provided for the integration of external antennas.

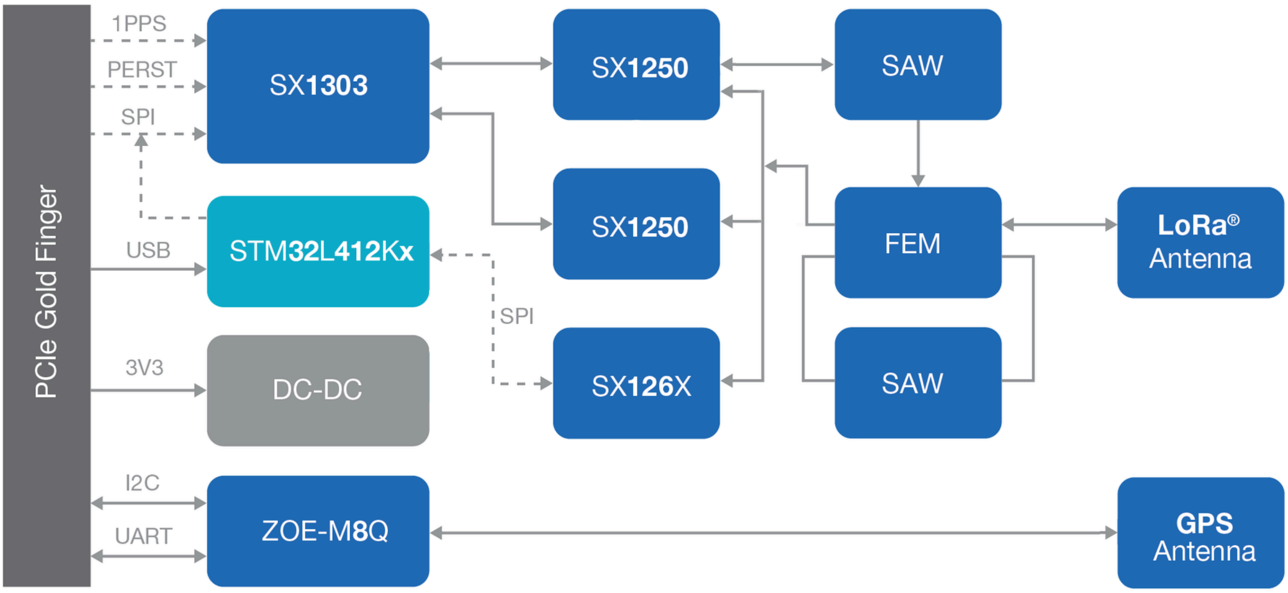


Figure 2: Block Diagram

Hardware

The hardware is categorized into seven (7) parts. It discusses the interfacing, pinouts, and its corresponding functions and diagrams. It also covers the parameters and standard values of the board.

Interfaces

Power Supply

The RAK5146 concentrator module must be powered through the 3.3 Vaux pins by a DC power supply. It is crucial that the voltage remains stable, as the current drawn can vary significantly during operation, depending on the power consumption profile of the SX1303 chip. For more detailed information, refer to the SX1303 Datasheet .

SPI Interface

SPI interface mainly provides for the Host_SCK, Host_MISO, Host_MOSI, Host_CSN pins of the system connector. The SPI interface gives access to the configuration register of SX1303 via a synchronous full-duplex protocol. Only the slave side is implemented.

USB Interface

The USB interface mainly provides for the USB_D+, USB_D- pins of the system connector. The USB interface gives access to the configuration register of SX1303 via an MCU STM32L412KBU6. Only the slave side is implemented.

UART and I2C Interface

The RAK5146 integrates a ZOE-M8Q GPS module, which supports both UART and I2C interfaces. The PINs on the golden finger provide a UART connection and an I2C connection, which allows direct access to the GPS module. The PPS (Pulse Per Second) signal is connected internally to the SX1303, as well as to the golden finger, making it available for use by the host board.

GPS_PPS

The RAK5146 includes the GPS_PPS input for received packets time-stamped and Fine timestamp.

RESET

The RAK5146 SPI card includes the RESET active-high input signal to reset the radio operations as specified by the SX1303 Specification. RAK5146 USB card's RESET is controlled by the MCU.

Antenna RF Interface

The module has one RF interface over a standard UFL connector (Hirose U. FL-R-SMT) with a characteristic impedance of 50 Ω. The RF port (J1) supports both Tx and Rx, providing the antenna interface.

Pin Definition

RAK**5146** SPI/USB PINOUT

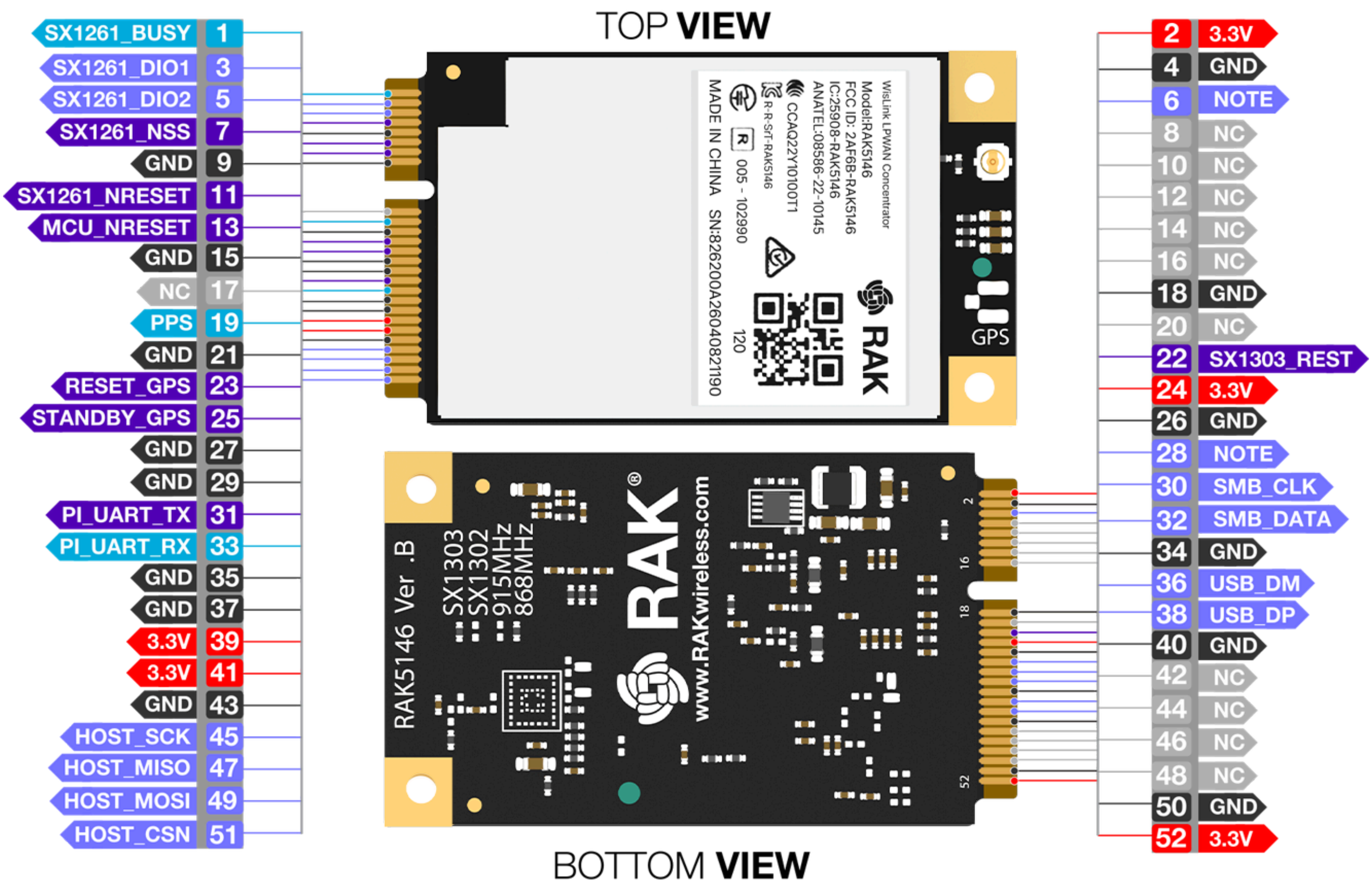


Figure 3: Pinout Diagram

Type	Description
IO	Bidirectional
DI	Digital input
DO	Digital output
OC	Open collector
OD	Open drain

Type	Description
PI	Power input
PO	Power output
NC	No connection

Pin No.	mPCIe Pin Rev. 2.0	RAK5146 Pin	Type	Description	Remarks
1	WAKE#	SX1261_BUSY	DO	No connection by default	Reserved for future applications
2	3.3 Vaux	3V3	PI	3.3 V_{DC} supply	
3	COEX1	SX1261_DIO1	IO	No connection by default	Reserved for future applications
4	GND	GND		Ground	
5	COEX2	SX1261_DIO2	IO	No connection by default	Reserved for future applications
6	1.5 V	GPIO(6)	IO	No connection by default	Connect to the SX1303's GPIO (6)
7	CLKREQ#	SX1261_NSS	DI	No connection by default	Reserved for future applications
8	UIM_PWR	NC		No connection	
9	GND	GND		Ground	

Pin No.	mPCIe Pin Rev. 2.0	RAK5146 Pin	Type	Description	Remarks
10	UIM_DATA	NC		No connection	
11	REFCLK-	SX1261_NRESET	DI	No connection by default	Reserved for future application
12	UIM_CLK	NC		No connection	
13	REFCLK+	MCU_NRESET	DI	RESET signal for MCU of RAK5146-USB	Active low
14	UIM_RESET	NC		No connection	
15	GND	GND		Ground	
16	UIM_VPP	NC		No connection	
17	RESERVED	NC		No connection	
18	GND	GND		Ground	
19	RESERVED	PPS	DO	Time pulse output	Leave open if not in use
20	W_DISABLE#	NC		No connection	
21	GND	GND		Ground	
22	PERST#	SX1303_RESET	DI	RAK5146-SPI reset input	Active high, ≥100 ns for SX1303 reset

Pin No.	mPCIe Pin Rev. 2.0	RAK5146 Pin	Type	Description	Remarks
23	PERn0	RESET_GPS	DI	GSP module ZOE-M8Q reset input	Active low, leave open if not in use
24	3.3Vaux	3v3	PI	3.3 V_{DC} supply	
25	PERp0	STANDBY_GPS	DI	GPS module ZOE-M8Q external interrupt input	Active low, leave open if not in use
26	GND	GND		Ground	
27	GND	GND		Ground	
28	1.5 V	GPIO(8)		Connect to the SX1303's GPIO (8)	
29	GND	GND		Ground	
30	SMB_CLK	I2C_SCL	IO	HOST SCL	Connect to GPS module ZOE-M8Q's SCL internally, leave open if not in use
31	PETn0	PI_UART_TX	DI	HOST UART_TX	Connect to GPS module ZOE-M8Q's UART_RT internally, leave open if not in use

Pin No.	mPCIe Pin Rev. 2.0	RAK5146 Pin	Type	Description	Remarks
32	SMB_DATA	I2C_SDA	IO	HOST SDA	Connect to GPS module ZOE-M8Q's SDA internally, leave open if not in use
33	PETp0	PI_UART_RX	DO	HOST UART_RX	` Connect to GPS module ZOE-M8Q's UART_TX internally, leave open if not in use
34	GND	GND		Ground	
35	GND	GND		Ground	
36	USB_D-	USB_DM	IO	USB differential data (-)	Require differential impedance of 90 Ω
37	GND	GND		Ground	
38	USB_D+	USB_DP	IO	USB differential data (+)	Require differential impedance of 90 Ω
39	3.3 Vaux	3V3	PI	3.3 V_{DC} supply	
40	GND	GND		Ground	
41	3.3Vaux	3V3	Pi	3.3 V_{DC} supply	
42	LED_WWAN#	NC		No connection	

Pin No.	mPCIe Pin Rev. 2.0	RAK5146 Pin	Type	Description	Remarks
43	GND	GND		Ground	
44	LED_WLAN#	NC		No connection	
45	RESERVED	HOST_SCK	IO	Host SPI SCK	
46	LED_WPAN#	NC		No connection	
47	RESERVED	HOST_MISO	IO	Host SPI MISO	
48	1.5 V	NC		No connection	
49	RESERVED	HOST_MOSI	IO	Host SPI MOSI	
50	GND	GND		Ground	
51	RESERVED	HOST_CSN	IO	Host SPI CSN	
52	3.3 Vaux	3V3	PI	3.3 V_{DC} supply	

LED Definition

Figure 4: RAK5146 LED Definition

RF Characteristics

Operating Frequencies

The board supports the following LoRaWAN frequency channels, allowing easy configuration while building the firmware from the source code.

Region	Frequency (MHz)
Europe	EU868 EU433
North America	US915
Russia	RU864
Asia	AS923
Australia	AU915
Korea	KR920

Region	Frequency (MHz)
India	IN865
China	CN470

RF Characteristics

The following table gives typically sensitivity level of the RAK5146 concentrator module.

Signal Bandwidth (kHz)	Spreading Factor	Sensitivity (dBm)
125	12	-139
125	7	-125
250	7	-123
500	12	-134
500	7	-120

Electrical Requirements

Exceeding any limits specified in the Absolute Maximum Ratings section can result in permanent damage to the device. These ratings are intended for stress testing only. Operating the module under these conditions, or outside the parameters defined in the Operating Conditions section, should be avoided. Extended exposure to Absolute Maximum Rating conditions may compromise the device's reliability.

The operating condition range defines the limits within which the functionality of the device is guaranteed. While application information is provided, it is advisory only and does not form part of the specification.

Absolute Maximum Rating

The limiting values given below are following the Absolute Maximum Rating System (IEC 134).

Symbol	Description	Condition	Min	Max
3.3Vaux	Module supply voltage	Input DC voltage at 3.3Vaux pins	-0.3 V	3.6 V
USB	USB D+/D- pins	Input DC voltage at USB interface pins	-	3.6 V
RESET	RAK5146 reset input	Input DC voltage at RESET input pin	-0.3 V	3.6 V
SPI	SPI interface	Input DC voltage at SPI interface pin	-0.3 V	3.6 V
GPS_PPS	GPS 1 PPS input	Input DC voltage at GPS_PPS input pin	-0.3 V	3.6 V
Rho_ANT	Antenna ruggedness	Output RF load mismatch ruggedness at ANT1	-	10:1 VSWR
Tstg	Storage temperature		-40° C	85° C

WARNING

The product is not protected against overvoltage or reversed voltages. If necessary, voltage spikes exceeding the power supply voltage specification, as outlined in the table above, must be limited to values within the specified boundaries by using appropriate protection devices.

Maximum ESD

The table below lists the maximum ESD.

Parameter	Min	Typical	Max	Remarks
ESD_HBM			1000 V	Charged Device Model JESD22-C101 CLASS III
ESD_CDM			1000 V	Charged Device Model JESD22-C101 CLASS III

NOTE

While this module is designed for robustness, it remains susceptible to damage from electrostatic discharge (ESD). Proper ESD protection must be maintained during handling or transportation. Static charges on the human body or equipment can easily reach several kilovolts and discharge undetectably. To prevent damage, always follow industry-standard ESD handling precautions.

Power Consumption

Mode	Condition	Min	Typical	Max
Active Mode (TX)	The power of the TX channel is 27 dBm and 3.3 V supply.	511 mA	512 mA	513 mA
Active Mode (RX)	TX disabled and RX enabled.	70 mA	81.6 mA	101 mA

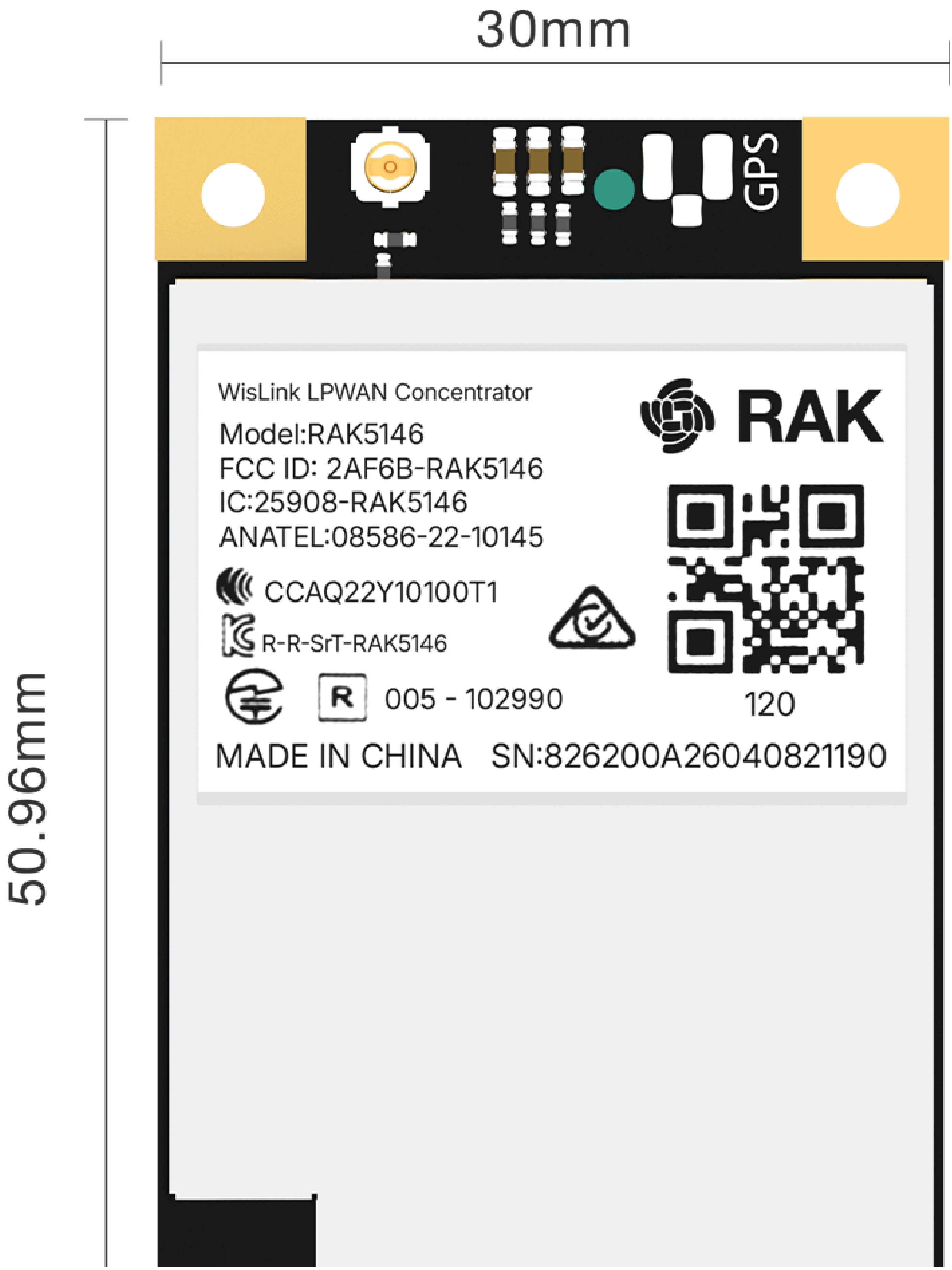
Power Supply Range

Input voltage at **3.3 Vaux** must be above the normal operating range minimum limit to switch-on the module.

Symbol	Parameter	Min	Typical	Max
3.3 Vaux	Module supply operating input voltage	3 V	3.3 V	3.6 V

Mechanical Characteristics

The board weighs 16.3 grams, is 30 mm wide, and 50.96 mm tall. The dimensions of the module fall completely within the **PCI Express Mini Card Electromechanical Specification**, except for the card's thickness (5.5 mm at its thickest).


30mm
50.96mm
GPS
WisLink LPWAN Concentrator
Model:RAK5146
FCC ID: 2AF6B-RAK5146
IC:25908-RAK5146
ANATEL:08586-22-10145
CCAQ22Y10100T1
R-R-SrT-RAK5146
R
005 - 102990
RAK
120
MADE IN CHINA
SN:826200A26040821190

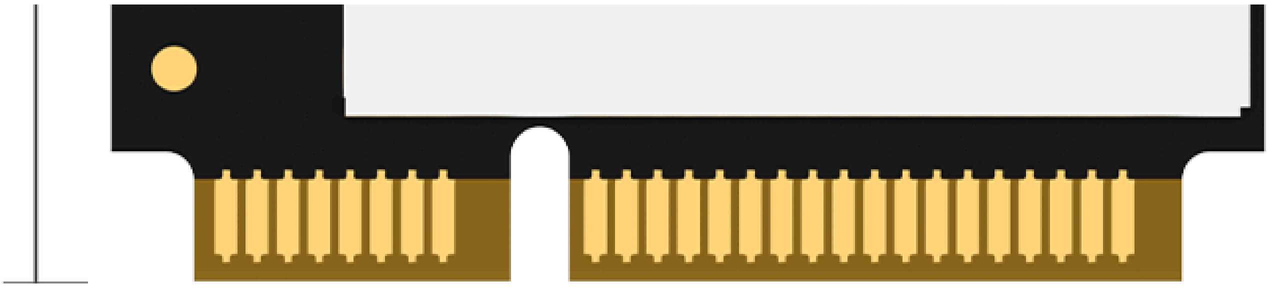


Figure 5: Dimensions

Environmental Requirements

Operating Conditions

Parameter	Min	Typical	Max	Remarks
Normal operating temperature	-40° C	+25° C	+85° C	Normal operating temperature range (fully functional and meet 3GPP specifications)

> **NOTE**
> Unless specified otherwise, all operating condition specifications are based on an ambient temperature of 25° C. Operation outside the specified conditions is not recommended, as extended exposure beyond these limits may impact the device's reliability.

Schematic Diagram

The RAK5146 concentrator module is based on Semtech's reference design for the SX1303. The SPI interface is available on the PCIe connector. **Figure 5** illustrates the minimum application schematic for the RAK5146 card. It should be powered with at least 3.3 V / 1 A DC power and have the SPI interface connected to the main processor.

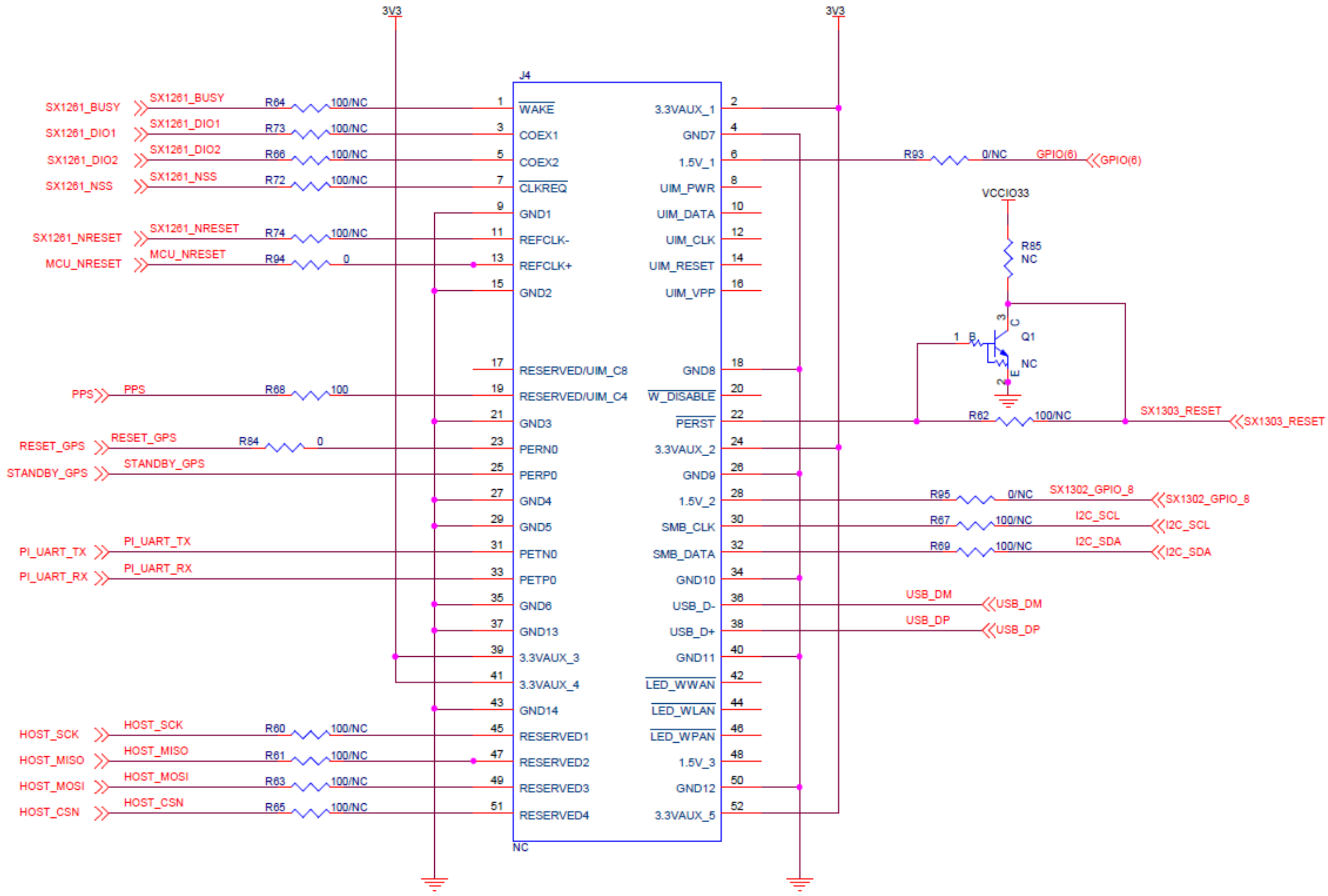


Figure 6: Schematic Diagram

Models / Bundles

In general, the RAK5146's variation is defined as **RAK5146 - XYZ**, where **X, Y, Z is the model variant.** Refer to the tables below to understand the different variants and their specifications.

Symbol	Description
X - Supported region	• 1 - US915 = 9XX MHz for US915/AU915/KR920/AS923 • 2 - EU868 = 8XX MHz for EU868/RU864/IN865
Y - Interface type	• 1 - SPI • 2 - USB

Symbol	Description
Z - Additional features	• 0 - No additional features • 2 - LBT • 5 - GPS • 6 - LBT+GPS

Model	Frequency	USB	SPI	LBT	GPS	PID
RAK5146-126	9XX MHz for US915/AU915/KR920/AS923	√		√	√	516013
RAK5146-122	9XX MHz for US915/AU915/KR920/AS924	√		√		516018
RAK5146-125	9XX MHz for US915/AU915/KR920/AS925	√			√	516011
RAK5146-120	9XX MHz for US915/AU915/KR920/AS926	√				516012
RAK5146-115	9XX MHz for US915/AU915/KR920/AS927		√		√	516010
RAK5146-110	9XX MHz for US915/AU915/KR920/AS928		√			516023
RAK5146-226	8XX MHz for EU868/RU864/IN865	√		√	√	515014
RAK5146-222	8XX MHz for EU868/RU864/IN866	√		√		515018

Model	Frequency	USB	SPI	LBT	GPS	PID
RAK5146-225	8XX MHz for EU868/RU864/IN867	√			√	515012
RAK5146-220	8XX MHz for EU868/RU864/IN868	√				515013
RAK5146-215	8XX MHz for EU868/RU864/IN869		√		√	515011
RAK5146-210	8XX MHz for EU868/RU864/IN870		√			515022

Certification